FIBRE PRODUCTS OF AFFINE FORMAL ALGEBRAIC SPACES OVER A SEPARATED BASE ARE AFFINE, AND MCQUILLAN IN THE MCQUILLAN CASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's own wording 'Assume X and Y corresponds to' and displays two chains of morphisms and one identification with a spectrum of a tensor product; all are reproduced verbatim.

Editorial note. This candidate was set aside earlier on the old gate's public-item collision verdict and re-admitted after the gate repair, the re-run being recorded in the queue.

Editorial note. The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Affine formal algebraic spaces as filtered colimits of affine schemes. Same chapter, Definition definition-affine-formal-algebraic-space, invoked at proof lines 3719-3721. Supplies the presentations whose fibre products are taken.
- The defining condition for a McQuillan affine formal algebraic space. Same chapter, Lemma lemma-mcquillan-affine-formal-algebraic-space, invoked at proof lines 3729-3732. Gives the second assertion once the closed immersion into the product is exhibited.
- The completed tensor product of weakly admissible rings and its topology. Same chapter, Definition definition-toplogy-tensor-product, invoked at proof lines 3738-3741. Identifies the ring of the fibre product as the completed tensor product.

Source reference. Original lemma 0AIC: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2135–2135. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
   (a) each X_λ is affine,
   (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

such that

$$X \cong \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Definition 0.2. Let R be a topological ring. Let M and N be linearly topologized R-modules. The *tensor product* of M and N is the (usual) tensor product $M \otimes_R N$ endowed with the linear topology defined by declaring

$$\operatorname{Im}(M_\mu \otimes_R N + M \otimes_R N_\nu \longrightarrow M \otimes_R N)$$

to be a fundamental system of open submodules, where $M_\mu \subset M$ and $N_\nu \subset N$ run through fundamental systems of open submodules in M and N. The *completed tensor product*

$$M\widehat{\otimes}_R N = \lim M \otimes_R N/(M_\mu \otimes_R N + M \otimes_R N_\nu) = \lim M/M_\mu \otimes_R N/N_\nu$$

is the completion of the tensor product.

Lemma 0.3. *Let S be a scheme. Let $X \to Z$ and $Y \to Z$ be morphisms of formal algebraic spaces over S. Assume Z separated.*

(1) *If X and Y are affine formal algebraic spaces, then so is $X \times_Z Y$.*
(2) *If X and Y are McQuillan affine formal algebraic spaces, then so is $X \times_Z Y$.*
(3) *If X, Y, and Z are McQuillan affine formal algebraic spaces corresponding to the weakly admissible topological S-algebras A, B, and C, then $X \times_Z Y$ corresponds to $A\widehat{\otimes}_C B$.*

Proof. Write $X = \operatorname{colim} X_\lambda$ and $Y = \operatorname{colim} Y_\mu$ as in Definition 0.1. Then $X \times_Z Y = \operatorname{colim} X_\lambda \times_Z Y_\mu$. Since Z is separated the fibre products are affine, hence we see that (1) holds. Assume X and Y corresponds to the weakly admissible topological S-algebras A and B and $X_\lambda = \operatorname{Spec}(A/I_\lambda)$ and $Y_\mu = \operatorname{Spec}(B/J_\mu)$. Then

$$X_\lambda \times_Z Y_\mu \to X_\lambda \times Y_\mu \to \operatorname{Spec}(A \otimes B)$$

is a closed immersion. Thus one of the conditions of Lemma 0AIC holds and we conclude that $X \times_Z Y$ is McQuillan. If also Z is McQuillan corresponding to C, then

$$X_\lambda \times_Z Y_\mu = \operatorname{Spec}(A/I_\lambda \otimes_C B/J_\mu)$$

hence we see that the weakly admissible topological ring corresponding to $X \times_Z Y$ is the completed tensor product (see Definition 0.2). □